Slide 1: PV101 Day 6

Troubleshooting Like a Professional

Slide 2: The expensive failure often has the cheapest cause

A hidden fuse can stop a whole system.

Slide 3: Today's rule

Stop. Look. Think. Measure. Verify. Then repair.

Slide 4: What worked yesterday can fail today

Your job is to prove where the power path opens.

Slide 5: Power flow review

Module -> controller -> battery -> protection -> inverter -> load.

Slide 6: Protection devices are part of the circuit

A fuse is not an accessory. It is a controlled weak point.

Slide 7: A fuse opens for a reason

Too much current, wrong rating, short circuit, damaged conductor, or installation error.

Slide 8: Do not bypass protection

Bypassing a fuse removes the safety feature and creates risk.

Slide 9: Meter first, screwdriver second

The meter tells you where voltage exists and where it stops.

Slide 10: Voltage tracing

Check source, input, output, and load side in order.

Slide 11: Continuity checks

Use continuity only when the circuit is de-energized.

Slide 12: Bad assumptions waste time

Do not blame the biggest part just because it is expensive.

Slide 13: Professional evidence

I measured voltage at the fuse input and zero volts at the fuse output.

Slide 14: Common PV system failures

Open fuse, loose terminal, wrong polarity, low battery, shaded module, failed connector.

Slide 15: Today's challenge

This system worked yesterday. Today it does not. Figure out why.

Slide 16: Team rules

No random part swapping. No fuse bypassing. Record every measurement.

Slide 17: Winning condition

Find it, explain it, repair it, demonstrate operation.

Slide 18: Debrief question 1

What did you check first and why?

Slide 19: Debrief question 2

What measurement finally told the truth?

Slide 20: The reveal

Each system had exactly one bad fuse.

Slide 21: Field lesson

Big equipment often depends on small protection devices.

Slide 22: Technician mindset

Verify before replacing.

Slide 23: Reflection

How will today's lesson change your troubleshooting?

Slide 24: Quote of the day

Professionals don't guess. Professionals verify.

Slide 25: Close

The cheapest part in the system can stop the entire system.